
UCL COMP0078: Supervised Learning

Problem Sheet

Features, Kernels and Regularisation

Solutions by Antonin Schrab

If you find any mistakes/typos, please contact sl-support@cs.ucl.ac.uk.

Overview of Problems

- Problem 1.1: Deriving results related to convexity (important to guarantee existence/uniqueness of a solution to an optimisation problem).
- Problem 1.2: Understanding when to use the primal or the dual solution to the linear regression optimisation problem.
- Problem 1.3 & 1.4: Proving whether some functions are valid kernels (if they are kernels they can then be used for any kernelised algorithm).
- Problem 1.5: Proving that the Gaussian kernel cannot admit a finite dimensional feature map, justifying the need for the use of infinite dimensional feature maps.
- Problem 2.1: Basic exercise on vector space definition.
- Problem 2.2: Example of a kernel defined on sets, highlighting the general use cases of kernels which are not restricted to the commonly used $\mathbb{R}^d$ space.
- Problem 2.3: Showing that deriving the (infinite) feature map of a kernel can be very complicated, this highlights the importance of the ‘kernel trick’ that kernels can simply be used without the need to derive their complicated corresponding feature map.

Definitions: Convexity

A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is convex if

$$f(\alpha p + (1 - \alpha)q) \leq \alpha f(p) + (1 - \alpha)f(q)$$

for all $p, q \in \mathcal{X}$ and all $\alpha \in (0, 1)$.

A set S is convex if for all $p, q \in S$ and all $\alpha \in (0, 1)$, we have $\alpha p + (1 - \alpha)q \in S$.

Problem 1.1.1

If f and g are convex then $f + g$ is convex.

Solution 1.1.1

For $p, q \in \mathcal{X}$ and $\alpha \in (0, 1)$, we have

$$\begin{aligned}(f + g)(\alpha p + (1 - \alpha)q) &= f(\alpha p + (1 - \alpha)q) + g(\alpha p + (1 - \alpha)q) \\ &\leq \alpha f(p) + (1 - \alpha)f(q) + \alpha g(p) + (1 - \alpha)g(q) \\ &= \alpha(f(p) + g(p)) + (1 - \alpha)(f(q) + g(q)).\end{aligned}$$

This shows that $f + g$ is convex.

Problem 1.1.2

If f is convex and g is affine (linear + a constant) then $f(g(\cdot))$ is convex.

Solution 1.1.2

Since g is affine it can be written as $g(x) = a(x + b) = ax + ab$ for some $a, b \in \mathbb{R}$ and for all $x \in \mathbb{R}$. We then have

$$\begin{aligned}
(f \circ g)(\alpha p + (1 - \alpha)q) &= f(g(\alpha p + (1 - \alpha)q)) \\
&= f(\alpha ap + (1 - \alpha)aq + ab) \\
&= f(\alpha a(p + b) + (1 - \alpha)a(q + b)) \\
&\leq \alpha f(a(p + b)) + (1 - \alpha)f(a(q + b)) \\
&= \alpha f(g(p)) + (1 - \alpha)f(g(q)).
\end{aligned}$$

This shows that $f \circ g$ is convex.

Problem 1.1.3

Suppose M is a symmetric matrix then M is a PSD matrix if and only if $f(\mathbf{x}) = \mathbf{x}^\top M \mathbf{x}$ is convex.

Solution 1.1.3

Suppose M is a symmetric PSD matrix. Note that the Hessian of f is

$$\frac{\partial^2}{\partial \mathbf{x}^2} f(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} \left(\frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top M \mathbf{x} \right) = \frac{\partial}{\partial \mathbf{x}} ((M + M^\top) \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} 2M \mathbf{x} = 2M$$

which is PSD since M is PSD. It follows from problem 1.1.5 (proved below) that f is convex. Now, suppose f is convex and so it attains its minimum at

$$0 = \frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top M \mathbf{x} = (M + M^\top) \mathbf{x} = 2M \mathbf{x}$$

using the Chain Rule. We deduce that f achieves its minimum at $\mathbf{0}$, so

$$\mathbf{x}^\top M \mathbf{x} = f(\mathbf{x}) \geq f(\mathbf{0}) = \mathbf{0}^\top M \mathbf{0} = 0.$$

This shows that M is positive semidefinite.

Problem 1.1.4

A level set of a convex function is convex.

Solution 1.1.4

Let f be a convex function. The level set of f at c is $L := \{x : f(x) \leq c\}$. Suppose $p, q \in L$, that is, $f(p) \leq c$ and $f(q) \leq c$. We then have

$$f(\alpha p + (1 - \alpha)q) \leq \alpha f(p) + (1 - \alpha)f(q) \leq \alpha c + (1 - \alpha)c = c,$$

so $\alpha p + (1 - \alpha)q \in L$. This proves that the level set L of f is a convex set.

Problem 1.1.5

For $f: \mathcal{X} \rightarrow \mathbb{R}$ if $\mathcal{X} \subseteq \mathbb{R}^n$ is a convex set and if the Hessian of f evaluated at $\mathbf{x}$ denoted $H(\mathbf{x})$ is positive semidefinite for all $\mathbf{x} \in \mathcal{X}$ then f is convex.

Solution 1.1.5

Let $\mathbf{x}, \mathbf{a} \in \mathcal{X}$. By Taylor's Theorem, there exists $t \in (0, 1)$ such that

$$f(\mathbf{x}) = f(\mathbf{a}) + (\mathbf{x} - \mathbf{a})^\top \nabla f(\mathbf{a}) + \frac{1}{2}(\mathbf{x} - \mathbf{a})^\top H(\mathbf{a} + t(\mathbf{x} - \mathbf{a}))(\mathbf{x} - \mathbf{a}).$$

Since $H(\mathbf{a} + t(\mathbf{x} - \mathbf{a}))$ is PSD by assumption, for $\mathbf{x} - \mathbf{a} \in \mathcal{X}$, we have

$$(\mathbf{x} - \mathbf{a})^\top H(\mathbf{a} + t(\mathbf{x} - \mathbf{a}))(\mathbf{x} - \mathbf{a}) \geq 0,$$

from which we deduce that

$$f(\mathbf{x}) \geq f(\mathbf{a}) + (\mathbf{x} - \mathbf{a})^\top \nabla f(\mathbf{a}).$$

Now, consider $\mathbf{p}, \mathbf{q} \in \mathcal{X}$, $\alpha \in [0, 1]$, and let

$$\mathbf{a} := \alpha \mathbf{p} + (1 - \alpha) \mathbf{q}$$

which belongs to $\mathcal{X}$ as $\mathcal{X}$ is a convex set. Note that

$$\mathbf{p} - \mathbf{a} = (1 - \alpha)(\mathbf{p} - \mathbf{q}) \quad \text{and} \quad \mathbf{q} - \mathbf{a} = \alpha(\mathbf{q} - \mathbf{p}).$$

We have

$$\begin{aligned} \alpha f(\mathbf{p}) + (1 - \alpha)f(\mathbf{q}) &\geq \alpha(f(\mathbf{a}) + (\mathbf{p} - \mathbf{a})^\top \nabla f(\mathbf{a})) + (1 - \alpha)(f(\mathbf{a}) + (\mathbf{q} - \mathbf{a})^\top \nabla f(\mathbf{a})) \\ &= f(\mathbf{a}) + \nabla f(\mathbf{a})^\top (\alpha(\mathbf{p} - \mathbf{a}) + (1 - \alpha)(\mathbf{q} - \mathbf{a})) \\ &= f(\mathbf{a}) + \nabla f(\mathbf{a})^\top (\alpha(1 - \alpha)(\mathbf{p} - \mathbf{q}) + (1 - \alpha)\alpha(\mathbf{q} - \mathbf{p})) \\ &= f(\mathbf{a}) \\ &= f(\alpha \mathbf{p} + (1 - \alpha) \mathbf{q}) \end{aligned}$$

which proves that f is convex.

Problem 1.2

Consider the solution to linear regression optimisation problem. When is it advantageous to compute it via the primal solution? When is it advantageous to compute it via the dual solution? Explain why.

Solution 1.2

Recall that the matrix X is of size $m \times n$ where we have m samples in $\mathbb{R}^n$. The primal solution is

$$\mathbf{w} = (X^\top X + \lambda I_n)^{-1} X^\top y$$

and the dual solution is

$$\mathbf{w} = \sum_{i=1}^m \alpha_i \mathbf{x}_i \quad \text{where} \quad \boldsymbol{\alpha} = (X X^\top + \lambda I_m)^{-1} y.$$

The operation with the highest time complexity in those formulas are the two matrix inversions. If we have more dimensions n than samples m , it is therefore advantageous to compute via the dual solution. For the case where we have more samples m than dimensions n , it is advantageous to use the primal solution.

Problem 1.3

Given a kernel $K: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ where $K(\mathbf{x}, \mathbf{t}) := (1 + \langle \mathbf{x}, \mathbf{t} \rangle)^2$. Find a feature map $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^6$ which corresponds to the kernel.

Solution 1.3

We have

$$\begin{aligned} K(\mathbf{x}, \mathbf{t}) &= (1 + \langle \mathbf{x}, \mathbf{t} \rangle)^2 \\ &= (1 + x_1 t_1 + x_2 t_2)^2 \\ &= 1 + x_1^2 t_1^2 + x_2^2 t_2^2 + 2x_1 t_1 + 2x_2 t_2 + 2x_1 x_2 t_1 t_2 \\ &= \begin{pmatrix} 1 \\ x_1^2 \\ x_2^2 \\ \sqrt{2}x_1 \\ \sqrt{2}x_2 \\ \sqrt{2}x_1 x_2 \end{pmatrix}^\top \begin{pmatrix} 1 \\ t_1^2 \\ t_2^2 \\ \sqrt{2}t_1 \\ \sqrt{2}t_2 \\ \sqrt{2}t_1 t_2 \end{pmatrix} \\ &=: \phi(\mathbf{x})^\top \phi(\mathbf{t}). \end{aligned}$$

Problem 1.4

For each of the following functions $K: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ argue whether they are a valid kernel (i.e. the kernel can be written as an inner product in some feature space) and when the answer is positive derive an associated feature map representation.

1. $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t}$ where $D := \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$
2. $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t}$ where $D := \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix}$
3. $K(\mathbf{x}, \mathbf{t}) = \exp(x_1 t_1)$ where x_1 is the first component of the vector $\mathbf{x}$ and, likewise, t_1 is the first component of the vector $\mathbf{t}$
4. $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top \mathbf{t} - (\mathbf{x}^\top \mathbf{t})^2$
5. Now prove that if $K: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is a given valid kernel then the following transformed kernels are also valid.
 - (a) $K(A\mathbf{x}, A\mathbf{t})$ where A is a given 2×2 matrix
 - (b) $f(\mathbf{x})K(\mathbf{x}, \mathbf{t})f(\mathbf{t})$ where f is a given real-valued function

Solution 1.4.1

This is a kernel and the feature map can be obtained as

$$\begin{aligned} K(\mathbf{x}, \mathbf{t}) &= \mathbf{x}^\top D \mathbf{t} \\ &= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} \\ &= x_1 t_1 + 5x_2 t_2 \\ &= \begin{pmatrix} x_1 \\ \sqrt{5}x_2 \end{pmatrix}^\top \begin{pmatrix} t_1 \\ \sqrt{5}t_2 \end{pmatrix} \\ &=: \phi(\mathbf{x})^\top \phi(\mathbf{t}). \end{aligned}$$

Solution 1.4.2

Note that, for $\mathbf{x} = (1, 0)^\top \in \mathbb{R}^2$ and $c = 1 \in \mathbb{R}$, we have

$$c^2 K(\mathbf{x}, \mathbf{x}) = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = -x_1^2 + 2x_1x_2 + 2x_1x_2 + 4x_2^2 = -1 < 0.$$

So the function K is not positive semidefinite, hence, it does not define a valid kernel.

Solution 1.4.3

Using the Taylor series of the exponential function $\exp(u) = \sum_{n=0}^{\infty} \frac{u^n}{n!}$, we obtain

$$\begin{aligned} K(\mathbf{x}, \mathbf{t}) &= \exp(x_1 t_1) \\ &= \sum_{n=0}^{\infty} \frac{x_1^n t_1^n}{n!} \\ &= \begin{pmatrix} 1 \\ x_1 \\ x_1^2/\sqrt{2!} \\ x_1^3/\sqrt{3!} \\ \vdots \end{pmatrix}^\top \begin{pmatrix} 1 \\ t_1 \\ t_1^2/\sqrt{2!} \\ t_1^3/\sqrt{3!} \\ \vdots \end{pmatrix} \\ &=: \phi(\mathbf{x})^\top \phi(\mathbf{t}). \end{aligned}$$

We deduce that K is a valid kernel.

Solution 1.4.4

Note that, for $\mathbf{x} = (\sqrt{2}, 0)^\top \in \mathbb{R}^2$ and $c = 1 \in \mathbb{R}$, we have

$$c^2 K(\mathbf{x}, \mathbf{x}) = 2 - 4 = -2 < 0.$$

So the function K is not positive semidefinite, hence, it does not define a valid kernel.

Solution 1.4.5.a

Consider $c_1, \dots, c_m \in \mathbb{R}$ and $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^2$. Letting $\tilde{\mathbf{x}}_i := A\mathbf{x}_i \in \mathbb{R}^2$ for $i = 1, \dots, m$. Since K is a kernel, we have

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j K(\tilde{\mathbf{x}}_i, \tilde{\mathbf{x}}_j) \geq 0$$

or equivalently

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j K(A\mathbf{x}_i, A\mathbf{x}_j) \geq 0.$$

This proves that $K(A\mathbf{x}, A\mathbf{t})$ is positive semidefinite and hence it is a valid kernel.

Solution 1.4.5.b

Consider $c_1, \dots, c_m \in \mathbb{R}$ and $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^2$. Let $\tilde{c}_i := f(\mathbf{x}_i)c_i \in \mathbb{R}$ for $i = 1, \dots, m$. Since K is a kernel, we have

$$\sum_{i=1}^m \sum_{j=1}^m \tilde{c}_i \tilde{c}_j K(\mathbf{x}_i, \mathbf{x}_j) \geq 0$$

or equivalently

$$\sum_{i=1}^m \sum_{j=1}^m c_i c_j f(\mathbf{x}_i) K(A\mathbf{x}_i, A\mathbf{x}_j) f(\mathbf{x}_j) \geq 0.$$

This proves that $f(\mathbf{x})K(\mathbf{x}, \mathbf{t})f(\mathbf{t})$ is positive semidefinite and hence it is a valid kernel.

Problem 1.5

Consider a Gaussian kernel function $K: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $K(\mathbf{x}, \mathbf{z}) := e^{-\|\mathbf{x}-\mathbf{z}\|^2}$, does there exist a finite-dimensional feature map representation? I.e., does there exist a $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ such that $K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$? Indicate an answer “yes” or “no” and provide an argument supporting your answer. [hard]

Solution 1.5

No.

Suppose for contradiction that there exist a finite-dimensional feature map representation, that is there exists $\phi: \mathbb{R}^n \rightarrow \mathbb{R}^d$ for some finite $d \in \mathbb{N}$ such that

$$e^{-\|\mathbf{x}-\mathbf{z}\|^2} = K(\mathbf{x}, \mathbf{z}) = \sum_{i=1}^d \phi_i(\mathbf{x})\phi_i(\mathbf{z})$$

Consider the points

$$\mathbf{x}_\ell := (\ell, 0, \dots, 0) \in \mathbb{R}^n$$

for $\ell \in \mathbb{N} \setminus \{0\}$. For $\ell \neq h$, the angle θ between $\phi(\mathbf{x}_\ell)$ and $\phi(\mathbf{x}_h)$ in $\mathbb{R}^d$ satisfies

$$\cos \theta = \frac{\phi(\mathbf{x}_\ell)^\top \phi(\mathbf{x}_h)}{(\phi(\mathbf{x}_\ell)^\top \phi(\mathbf{x}_\ell))(\phi(\mathbf{x}_h)^\top \phi(\mathbf{x}_h))} = \frac{K(\mathbf{x}_\ell, \mathbf{x}_h)}{K(\mathbf{x}_\ell, \mathbf{x}_\ell)K(\mathbf{x}_h, \mathbf{x}_h)} = e^{-|\ell-h|} \leq e^{-1}$$

Since arccos is a decreasing function on $[-1, 1]$, we obtain that the angle satisfies

$$\theta \geq \arccos(e^{-1}) > 0.$$

This means that we have infinitely many vectors $\{\phi(\mathbf{x}_\ell) : \ell \in \mathbb{N} \setminus \{0\}\}$ in $\mathbb{R}^d$ with the angle between any pair of these being greater than $e^{-1} > 0$. This is a contradiction as there exist only finitely many such vectors (see next two paragraphs). Having reached a contradiction, we deduce that our hypothesis that there exists a finite-dimensional feature map was wrong. We have proved that there does not exist such a finite-dimensional feature map.

Consider a set of vectors $\mathcal{S}_\epsilon$ in $\mathbb{R}^d$, satisfying the property that the angle between any two vectors in $\mathcal{S}_\epsilon$ is greater than some positive constant $\epsilon > 0$. The length of the vectors does not influence the

angles between them, hence we can assume that they are of unit lengths, so the vectors lie on the d -dimensional unit sphere. The fact that $\mathcal{S}_\epsilon$ must be finite is intuitive (at least for $d = 2$ and $d = 3$) and can be assumed, we prove it for completeness.

Consider the d -dimensional unit sphere. Since it is contained within $[-1, 1]^d$, the volume of this sphere is bounded by 2^d (i.e. the volume of $[-1, 1]^d$). Now consider d vectors $\mathbf{v}_1, \dots, \mathbf{v}_d \in \mathcal{S}_\epsilon$ lying on the unit sphere. Since the angles between any two such vectors is greater than $\epsilon > 0$, the polytope with vertices $\{\mathbf{v}_1, \dots, \mathbf{v}_d, \mathbf{0}\}$ in $\mathbb{R}^d$ must have volume greater than some positive value V_ϵ depending on ϵ . So, d vectors in $\mathcal{S}_\epsilon$ contribute at least volume V_ϵ . Using a crude upper bound, $d \left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right)$ vectors in $\mathcal{S}_\epsilon$ form at least $\left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right)$ non-overlapping polytopes constructed as above. Those contribute at least volume $\left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right) V_\epsilon$ which is larger than the upper bound 2^d on the volume of the unit sphere. This is a contradiction since by adding volumes of non-overlapping polytopes contained within the unit sphere, we obtain larger volume than the volume of the sphere itself. We deduce that $\mathcal{S}_\epsilon \leq d \left(\left\lceil \frac{2^d}{V_\epsilon} \right\rceil + 1 \right) < \infty$.

Problem 2.1

Argue that the vector space definition implies that every element $\mathbf{x} \in X$ has an additive inverse $-\mathbf{x} \in X$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$.

Solution 2.1

We have

$$\mathbf{0} = 0\mathbf{x} = (1 - 1)\mathbf{x} = 1\mathbf{x} + (-1)\mathbf{x} = \mathbf{x} + (-1)\mathbf{x}$$

We deduce that the additive inverse of $\mathbf{x} \in X$ is $(-1)\mathbf{x} \in X$ which we simply write as $-\mathbf{x} \in X$.

Problem 2.2

Kernels between sets. Let X be a finite set. Define $K: 2^X \times 2^X \rightarrow \mathbb{R}$ as

$$K(A, B) := 2^{|A \cap B|}$$

where $A, B \subseteq X$. Prove that K is a kernel.

Solution 2.2

Recall that 2^X is the power set of X (i.e. the set of all subsets of X , including the empty set), and that $|2^X| = 2^{|X|}$.

To prove that K is a kernel, we construct a feature map. For a statement S which is either true or false, we use the notation

$$[S] = \begin{cases} 1 & \text{if } S \text{ is true,} \\ 0 & \text{if } S \text{ is false.} \end{cases}$$

Now, define the features

$$\phi_C(A) = [C \subseteq A]$$

indexed by subsets $C \in 2^X$ (as opposed to the more conventional indices $i = 1, 2, 3, \dots$). For $A, B \in 2^X$, we then have

$$\begin{aligned} \sum_{C \in 2^X} \phi_C(A) \phi_C(B) &= \sum_{C \in 2^X} [C \subseteq A][C \subseteq B] = \sum_{C \in 2^X} [C \subseteq A \cap B] \\ &= \sum_{C \in 2^{A \cap B}} 1 + \sum_{C \in 2^X \setminus 2^{A \cap B}} 0 = |2^{A \cap B}| = 2^{|A \cap B|} = K(A, B). \end{aligned}$$

This proves that K is indeed a kernel.

Problem 2.3

Min kernel.

1. Argue that $\min(x, t)$ (where $x, t \in [0, \infty)$) is a kernel. [technical]
2. Let $M > 0$. Find a set of basis functions $\phi_\ell : [0, M] \rightarrow \mathbb{R}$ for $\ell \in \mathbb{N}$ such that

$$\min(x, t) = \sum_{\ell=0}^{\infty} \phi_\ell(x) \phi_\ell(t).$$

[technical, very difficult]

Solution 2.3.1

Consider the inner product space $\mathcal{F}$ which consists of functions from $[0, \infty)$ to $\mathbb{R}$ satisfying

1. $f(0) = 0$
2. f is absolutely continuous (hence $f(b) - f(a) = \int_a^b f'(u) du$)
3. $\int_0^\infty (f'(u))^2 du < \infty$

with the inner product

$$\langle f, g \rangle := \int_0^\infty f'(u) g'(u) du$$

for $f, g \in \mathcal{F}$.

We consider the feature map $\phi : [0, \infty) \rightarrow \mathcal{F}$ defined as $\phi(x) := \Phi_x \in \mathcal{F}$ for all $x \geq 0$, where

$$\Phi_x(u) := \begin{cases} u & \text{if } u \leq x, \\ x & \text{if } u > x, \end{cases}$$

for all $u \geq 0$. It can easily be verified that $\Phi_x \in \mathcal{F}$ for all $x \geq 0$ (i.e. that it satisfies the three conditions). Note that, when we differentiate Φ_x for some $x \geq 0$, we obtain the step-function

$$\Phi'_x(u) = \begin{cases} 1 & \text{if } u \leq x, \\ 0 & \text{if } u > x, \end{cases}$$

for all $u \geq 0$. Note that, for $x, t \in \mathbb{R}$, we have

$$\Phi'_x(u) \Phi'_t(u) = \begin{cases} 1 & \text{if } u \leq \min(x, t), \\ 0 & \text{if } u > \min(x, t), \end{cases}$$

for all $u \geq 0$. Finally, for $x, t \in \mathbb{R}$, we have

$$\langle \phi(x), \phi(t) \rangle = \langle \Phi_x, \Phi_t \rangle = \int_0^\infty \Phi'_x(u) \Phi'_t(u) du = \int_0^{\min(x, t)} 1 du + \int_{\min(x, t)}^\infty 0 du = \min(x, t)$$

This proves that $\min(x, t)$ is a kernel.

Solution 2.3.2

We work with the kernel $k(x, t) := \min(x, t)$ for $x, t \in [0, M]$. Consider the integral operator $T_k : L^2[0, M] \rightarrow L^2[0, M]$ which for $f \in L^2[0, M]$, is defined as

$$(T_k f)(x) := \int_0^M k(x, u) f(u) du.$$

Mercer's Theorem states that, $x, t \in [0, M]$, we have

$$k(x, t) = \sum_{\ell=0}^{\infty} \lambda_{\ell} e_{\ell}(x) e_{\ell}(t)$$

where $\{e_{\ell}\}_{\ell=0}^{\infty}$ is an *orthonormal basis* for $L^2[0, M]$ consisting of eigenfunctions of the operator T_k with associated eigenvalues $\{\lambda_{\ell}\}_{\ell=0}^{\infty}$. Note that, letting

$$\phi_{\ell} := \sqrt{\lambda_{\ell}} e_{\ell} \in L^2[0, M]$$

for $\ell \in \mathbb{N}$, we then have

$$\sum_{\ell=0}^{\infty} \phi_{\ell}(x) \phi_{\ell}(t) = \sum_{\ell=0}^{\infty} \sqrt{\lambda_{\ell}} e_{\ell}(x) \sqrt{\lambda_{\ell}} e_{\ell}(t) = \sum_{\ell=0}^{\infty} \lambda_{\ell} e_{\ell}(x) e_{\ell}(t) = k(x, t)$$

for all $x, t \in [0, M]$. Hence, if we obtain the eigenvalues and eigenfunctions of the operator T_k , we can recover the feature map of the kernel k .

We now explicitly derive the eigenpairs for the kernel $k(x, t) := \min(x, t)$ for $x, t \in [0, M]$. An eigenpair (λ, e) is by definition a scalar $\lambda \in \mathbb{R}$ and a function $e \in L^2[0, M]$ satisfying for all $x \in [0, M]$

$$\begin{aligned} \lambda e(x) &= (T_k e)(x) \\ &= \int_0^M \min(x, t) e(t) dt \\ &= \int_0^x \min(x, t) e(t) dt + \int_x^M \min(x, t) e(t) dt \\ &= \int_0^x t e(t) dt + \int_x^M x e(t) dt \end{aligned}$$

Letting $x = 0$, we get $\lambda e(0) = 0$ and deduce that $e(0) = 0$. We now differentiate both sides with respect to x using Leibniz integral rule and the Chain rule, we obtain

$$\begin{aligned} \lambda e'(x) &= \left(\frac{\partial}{\partial x} \int_0^x t e(t) dt \right) + x \frac{\partial}{\partial x} \left(\int_x^M e(t) dt \right) + \left(\frac{\partial x}{\partial x} \right) \left(\int_x^M e(t) dt \right) \\ &= x e(x) + x(-e(x)) + 1 \int_x^M e(t) dt \\ &= \int_x^M e(t) dt \end{aligned}$$

Letting $x = M$, we get $\lambda e'(M) = 0$ and deduce that $e'(M) = 0$. Differentiating with respect to x and using Leibniz integral rule one more time, we get

$$\lambda e''(x) = -e(x).$$

This is a second order differential equation which has solution

$$e(x) = c_1 \cos\left(\frac{x}{\sqrt{\lambda}}\right) + c_2 \sin\left(\frac{x}{\sqrt{\lambda}}\right)$$

for some constants $c_1, c_2 \in \mathbb{R}$. Note that $c_1 = e(0) = 0$ and

$$0 = e'(M) = \frac{c_2}{\sqrt{\lambda}} \cos\left(\frac{M}{\sqrt{\lambda}}\right)$$

which has solutions

$$\frac{M}{\sqrt{\lambda_\ell}} = \left(\ell + \frac{1}{2}\right) \pi \quad \Longleftrightarrow \quad \lambda_\ell = \frac{M^2}{\left(\ell + \frac{1}{2}\right)^2 \pi^2}$$

for $\ell \in \mathbb{N}$. We have

$$e_\ell(x) = c_2 \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right)$$

for $\ell \in \mathbb{N}$, since we want an orthonormal basis, we require

$$1 = \|e_\ell\|_2^2 = \int_0^M e_\ell(x)^2 dx = c_2^2 \int_0^M \sin^2\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right) dx = c_2^2 \frac{M}{2} \quad \Longleftrightarrow \quad c_2 = \sqrt{\frac{2}{M}}.$$

For $\ell \in \mathbb{N}$, the eigenpairs are then

$$\lambda_\ell = \frac{M^2}{\left(\ell + \frac{1}{2}\right)^2 \pi^2} \quad \text{and} \quad e_\ell(x) = \sqrt{\frac{2}{M}} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right)$$

which give the feature map

$$\phi_\ell(x) := \sqrt{\lambda_\ell} e_\ell(x) = \frac{\sqrt{2M}}{\left(\ell + \frac{1}{2}\right) \pi} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right).$$

for $x \in [0, M]$ We have proved that

$$k(x, t) = \sum_{\ell=0}^{\infty} \phi_\ell(x) \phi_\ell(t) = \sum_{\ell=0}^{\infty} \frac{2M}{\left(\ell + \frac{1}{2}\right)^2 \pi^2} \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} x\right) \sin\left(\frac{\left(\ell + \frac{1}{2}\right) \pi}{M} t\right)$$

for all $x, t \in [0, M]$.

To illustrate that this is the correct answer, we plot in Figure 1 the partial sum (replacing $\sum_{\ell=0}^{\infty}$ with $\sum_{\ell=0}^n$) for $n = 0, 1, 2, 3, 4$ with $M = 2$ and $t = 1$.

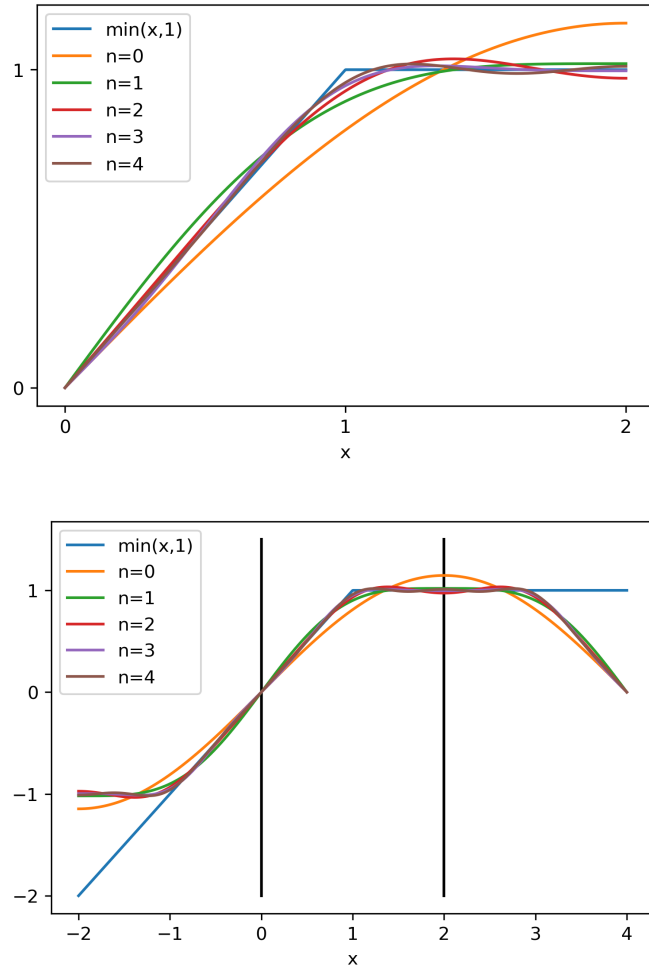


Figure 1: Min kernel partial sums.